

Grounded Agents: If Humanoids Cost Like Phones

Application Paper III of the *Grounded Agents* series
A Stress Test of the Framework’s Conditional Predictions

Macharia Bari

Abstract

We subject the industrial claims of the grounded-agents framework to their sharpest challenge: the scenario in which humanoid robots reach consumer electronics prices and become as easy to set up as a phone. We formalize the challenge as a conjunction of four conditions—hardware commoditization, automatic calibration, plug-and-play loop closure, and broad grounding transfer—and separate the framework’s conditional predictions into those that fall if the conditions hold, those that survive, and those that were never conditioned on integration cost in the first place. We then formalize the two transitions embedded in the phone analogy. The first is the microchip moment: integration of sensing, prediction, falsification, revision, and persistence into a reusable loop stack. The second is the internet moment: interoperability of loop protocols across domains. We prove that the value layer, the non-purchasability of grounding, the unpredictability of winners, and the commoditization of the model layer are invariant under the phone scenario, and we derive the strategic prescription that survives it: design for the standard, own the data that survives it. All results are conditional on the axioms of Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics*.

Remark 0.1 (Reading). Theorem numbers in this paper are local. The framework, definitions, and proofs of the results recalled in Section 2 live in Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics*; the industrial-structure claims tested here are from Application Paper I; the entry-point and transfer analysis used below is from Application Paper II.

Contents

1	Introduction	2
2	Recalled Framework	2
2.1	Concepts Imported from Paper 0, <i>The Inference-First Picture of Mind, Logic, and Mathematics</i>	2
2.2	Predictions Imported from Application Paper I and Application Paper II	3
3	The Phone Scenario	4
3.1	The Claim	4
3.2	What Would Have to Be True	4
4	What Falls and What Survives	5
4.1	What Falls	5
4.2	What Survives	5
5	The Two Transitions	6
5.1	The Cast of the Standardized Era	7

6	Strategy: The Fork	8
7	The Revised Prediction	9
8	Open Problems	10
9	Conclusion	11
A	Summary of Definitions	12
B	Summary of Theorems	12
C	A Note on Method	12

1 Introduction

Every structural theory should be set against its hardest counterexample, not its most convenient dataset. This paper takes the framework of Application Paper I and Application Paper II and attempts to break it. The counterexample: humanoid robots cost \$2,000 and set up like iPhones.

The industrial story of the framework begins with a claim about cost dynamics. Integration cost—the cost of closing an epistemic loop in an environment—decays more slowly than component cost. This single asymmetry carries the picture that follows: a staggered set of domain-specific eras, a loose federation of domain integrators, transfer as the generalization problem, and a consumer robotics market that is specialized long before it is general. If a humanoid can be unboxed and know a kitchen the way a phone knows a network, then either integration cost collapsed, or the loop standardized, or grounding transferred broadly—and all three falsify a different slice of the argument.

The paper proceeds in five parts. Section 2 recalls the framework’s machinery. Section 3 formalizes the phone scenario and asks what would have to be true. Section 4 separates what falls from what survives, and proves the invariance of the framework’s core claims. Section 5 formalizes the two historical transitions the analogy compresses, and works through who the Novells, Ciscos, Microsofts, Googles, and Apples of the loop layer would be. Section 6 derives the revised strategy and the revised prediction. We close with open problems.

A note on what this paper is testing. The framework is not a predictive oracle. It is a conditional theory. Its claims have the form: *if the antecedent holds, the consequent follows*. A stress test of such a theory is not the attempt to show one of its consequents is empirically false. It is the attempt to show that an antecedent the theory relies on can be negated, and to trace honestly which consequents fall with it. That is what this paper does.

2 Recalled Framework

2.1 Concepts Imported from Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics*

We use the framework developed in Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics* without proof. Full definitions and proofs are there.

Definition 2.1 (Agent and Grounding). An *agent* is a tuple $\mathcal{A} = (\mathcal{I}, \mathcal{S}, \mu)$ with inference system $\mathcal{I} = (\mathcal{C}, \vdash, \sigma)$, sensory states $\mathcal{S}$, and grounding map μ . The *grounding coefficient* $G(\mathcal{A})$ measures

the fraction of belief contents grounded in $\mathcal{A}$'s own sensory history, via the inferential closure of $\mathcal{A}$'s self-sourced contents.

Definition 2.2 (Environment-Indexed Grounding). The *environment-indexed grounding coefficient* is $G_{\mathcal{E}}(\mathcal{A}) = \inf_{s \in \mathcal{S}_{\mathcal{E}}} G(\mathcal{A}, s)$, over sensory states produced by interaction with $\mathcal{E}$.

Definition 2.3 (Loop Closure). Agent $\mathcal{A}$ *closes the loop* in $\mathcal{E}$ if it has a self-sourced observation operator with respect to $\mathcal{E}$ and $G_{\mathcal{E}}(\mathcal{A}) > 0$. Its *grounded data* is the set of belief contents in $\mathcal{E}$ not derivable from public corpora.

Theorem 2.4 (Grounding Cannot Be Purchased). *No purchase of components yields $G_{\mathcal{E}}(\mathcal{B}) > 0$ for an agent $\mathcal{B}$ without $\mathcal{B}$'s own interaction with $\mathcal{E}$.*

Theorem 2.5 (The Model Layer Commoditizes). *The generative components of the agent—the model layer—are not where the moat lives. Substituting one generative component for another leaves G invariant and cannot move an agent across the grounded/ungrounded threshold. The model layer is the ISA of the era, not the product.*

Theorem 2.6 (The Value Layer Shifts). *The value layer of the AI industry shifts structurally from the model layer to the loop layer. Whoever owns the grounded data owns the moat. This result depends only on grounding being the source of value, not on integration cost.*

Theorem 2.7 (Staggered Eras). *General-purpose robotics arrival is bounded below by the specialization arrivals; the robotics era is a sequence of domain-specific eras rather than a single event.*

2.2 Predictions Imported from Application Paper I and Application Paper II

The following claims are proved in Application Paper I and Application Paper II and tested in this paper. We restate them in the abstract form used below.

Claim	Source	Shorthand
Integration cost decays slower than component cost	Application Paper I	IC
Domain-specific integrators own each era	Application Paper I	INT
Loop standardization is the bottleneck	Application Paper I	LS
The tinkering threshold is delayed by generality	Application Paper I	TK
The winner is unpredictable (IBM pattern)	Application Paper I	IBM
Entry narrow, expand along max T	Application Paper II	ENT
T decides the industrial shape	Application Paper II	T
Structural inversion to the loop layer	Application Paper I, Application Paper II	INV

Remark 2.8 (The Anchoring Claim). All of the framework’s industrial timeline claims are conditioned, directly or indirectly, on IC: integration cost decays slower than component cost. If IC is false, then Application Paper I’s phase structure breaks. The phone scenario is precisely an attack on IC. This remark is the load-bearing observation of the paper.

3 The Phone Scenario

3.1 The Claim

Definition 3.1 (The Phone Scenario P). The *phone scenario* is the conjunction of four conditions on a humanoid robot H in environments $\mathcal{E}$:

- (P1) **Hardware commoditization.** $c_{\text{hw}}(H)$ falls to consumer electronics scale for actuators, sensors, compute, batteries, and materials.
- (P2) **Automatic calibration.** H calibrates its own sensors, learns its own kinematics, and adapts to its own body without human engineers.
- (P3) **Plug-and-play loop closure.** H closes its epistemic loop in a new environment $\mathcal{E}$ without custom integration. Equivalently: the loop standardizes the way TCP/IP standardized networking.
- (P4) **Broad grounding transfer.** H grounds itself in kitchens, workshops, offices, and outdoors without domain-specific data. Equivalently: $T > 1$ across almost all environment pairs.

Definition 3.2 (Antecedent Negation). A prediction is *falsified by* P if its negation follows from P together with the axioms of Paper 0, *The Inference-First Picture of Mind, Logic, and Mathematics*. A claim is *invariant under* P if it remains a consequence of the axioms when P is assumed.

3.2 What Would Have to Be True

Proposition 3.3 (Conditions (P1) and (P2) Are Engineering Problems). *Conditions (P1) and (P2) of Definition 3.1 are visible engineering trajectories: hardware commoditization is underway in Chinese manufacturing, and automatic calibration (self-calibrating sensors, learned kinematics, adaptive control) is an active research area with steady progress. Neither requires a change in the framework’s assumptions.*

Proposition 3.4 (Conditions (P3) and (P4) Are Framework Challenges). *Conditions (P3) and (P4) are the ones that conflict with the framework.*

Proof. Condition (P3) requires the epistemic loop to standardize across environments. By the vertical-specificity result of Application Paper I, grounding is environment-specific; standardization of the *interface* is possible, but standardization of the *domain-specific calibration* is not automatic. Condition (P4) requires $T > 1$ across almost all environment pairs, which by Application Paper II (transfer is structural, not architectural) requires near-total overlap in physics, sensory profile, causal structure, and object ontology across all pairs of environments. Both conditions demand the strongest forms of the framework’s weakest assumptions. $\square$

Proposition 3.5 (The Scenario as an Attack on IC). *The phone scenario P implies the negation of IC: if the loop is plug-and-play and transfer is broad, then integration cost has collapsed to the scale of component cost, and the anchoring claim fails.*

Proof. IC states that integration cost decays slower than component cost. Condition (P3) states that loop closure requires no custom integration, so c_{int} vanishes as a scaling factor. Condition (P1) states that c_{hw} reaches consumer scale. Hence total loop cost $c_{\text{loop}} = c_{\text{hw}} + c_{\text{env}} + c_{\text{int}}$ is driven by hardware alone, and integration cost no longer dominates the tinkering threshold. This is the negation of IC. $\square$

4 What Falls and What Survives

4.1 What Falls

Theorem 4.1 (Falsification Under P). *Assume the phone scenario P. Then:*

- (1) *IC is falsified (integration cost collapsed).*
- (2) *TK is falsified (general-purpose robots reach the tinkering threshold on the same timeline as specialized ones).*
- (3) *STG is falsified: the robotics era is a single event, like the PC era.*
- (4) *LS is resolved: loop closure has standardized.*

Proof. (1) is Proposition 3.5. (2) follows because the tinkering threshold in Application Paper I is governed by integration cost; if that cost collapses (condition (P3)), generality no longer delays affordability, so TK’s antecedent (integration cost decays slower than component cost) is false. (3) By Theorem 2.7, the era is staggered because grounding is environment-specific. Condition (P4) asserts broad transfer, i.e. the failure of environment-specificity; so the basis for staggering fails and the era collapses into a single event. (4) is condition (P3) itself. $\square$

Corollary 4.2 (INT Is Weakened). *If P holds, the domain-specific-integrator prediction INT is weakened: if $T > 1$ across the board, one loop generalizes everywhere, and the landscape could be one firm or a few winners rather than a federation of domain integrators.*

Proof. INT rests on vertical specificity, which condition (P4) negates. If a single loop generalizes everywhere, the moats that INT needs are unified, and the federated landscape collapses into a concentrated one. $\square$

Remark 4.3 (The Anticipated Failure Mode). The falsifications in Theorem 4.1 are not surprises internal to the framework. They are the “Integration Standardization” failure mode flagged in Application Paper I (the remark *Integration Standardization*): the framework anticipated that loop closure might standardize. It assigned that possibility low probability, not zero. The phone scenario is the low-probability branch being realized.

4.2 What Survives

Theorem 4.4 (Core Invariance Under P). *Assume the phone scenario P. The following claims remain consequences of the axioms of Paper 0, The Inference-First Picture of Mind, Logic, and Mathematics:*

- (1) ***The value layer still shifts.*** *The shift to the loop layer is independent of integration cost.*
- (2) ***Grounded data is still non-derivable and non-purchasable.*** *Grounding cannot be bought; the loop must be lived.*

(3) ***The winner is still unpredictable.** Even if the era arrives faster and more unified, the occupant of the value layer is not determined by the structure.*

(4) ***The model layer still commoditizes.** The model layer remains the ISA, not the product.*

Proof. (1) By Theorem 2.6, the value layer shifts because grounding is the source of value. Neither the direction nor the existence of the shift depends on integration cost; the shift’s speed and shape do, but not its occurrence. (2) By Theorem 2.4, grounding requires the agent’s own interaction with the environment. Hardware price (P1) does not negate this: cheap hardware is still hardware, and purchase of it is not interaction. (3) By the IBM-pattern result of Application Paper I, the structure predicts positions, not occupants. A faster, more unified era has the same property: occupancy is a matter of execution, timing, and contingency. (4) By Theorem 2.5, the model layer commoditizes because substitution of generative components leaves G invariant—a result with no dependence on integration cost. $\square$

Corollary 4.5 (The Core Insight Survives the Counterfactual). *Grounding is the moat, not the model. This claim is invariant under the phone scenario. What changes under P is the speed and shape of the transition, not its direction.*

Proof. Corollary 4.5 is the conjunction of items (1) and (4) of Theorem 4.4. $\square$

Remark 4.6 (What the Invariance Does Not Assert). Theorem 4.4 does not assert that the framework’s timeline predictions survive P. It asserts that the framework’s core structural claims survive. The moral of the section is the separation: the framework’s claims about *where value lives* are conditional only on grounding; the framework’s claims about *how quickly and in what shape* are conditional on integration cost. The phone scenario attacks the second class. It does not touch the first.

5 The Two Transitions

The phone analogy compresses two historical transitions into one phrase. Separating them is not pedantry; each transition has a distinct equivalent in the loop layer, and each must occur for the phone scenario to play out fully.

Definition 5.1 (The Two Transitions). *Transition 1, the microchip moment:* integration of components onto a single substrate. Its loop-layer equivalent is integration of sensing, prediction, falsification, revision, and persistence into a reusable loop stack—the loop-on-a-chip. *Transition 2, the internet moment:* interoperability of standard across networks. Its loop-layer equivalent is an interoperable loop protocol and a capability marketplace, allowing loops closed in one domain to invoke loops closed in another.

Historical era	What it solved	AI/robotics equivalent	Who rules
Pre-microchip (1950s–60s)	Discrete components, bespoke integration	Custom robot stacks, hand-calibrated loops	Labs, integrators, IBM-like firms
Microchip invention (1958–71)	Integration of components onto silicon	Integration of sensing, prediction, falsification, revision, persistence into a reusable loop stack	Whoever owns the loop-on-a-chip standard
Ethernet without internet (1980s)	Local networks, proprietary protocols	Domain-specific loops, no cross-domain standard	Novell-like domain integrators
Internet / TCP/IP (1983–95)	Interoperability across networks	Interoperable loop protocol, capability marketplace	Platforms, standard owners, data indexers

Proposition 5.2 (We Have Neither Transition). *If the phone path is to be followed, both transitions are required in the loop layer. We have neither yet. Hence the present period is the Novell era of robotics: many proprietary local loops, no interoperable standard, and domain-specific integrators ruling their own local networks.*

Proof. By Definition 5.1, the phone scenario’s full realization requires plug-and-play loop closure (condition (P3)) and broad transfer (condition (P4)), which are the loop-layer microchip moment and the loop-layer internet moment respectively. By Propositions 3.3 and 3.4, these are the conditions least advanced: loop closure remains domain-specific and T does not exceed 1 across modality clusters. Hence the loop layer’s equivalents of microchip and internet standardization are absent, and the era is characterized by the first of the three intermediate columns of the table: proprietary local loops. $\square$

5.1 The Cast of the Standardized Era

Proposition 5.3 (Who Rules Each Gate of the Standardized Loop Layer). *If both transitions occur, the rules of the loop layer accrue to distinct archetypes:*

- (1) **The Novells** are the domain integrators who rule a domain with proprietary loop architecture while the standard does not exist.
- (2) **The Cisco** supplies the infrastructure for the standard loop protocol—the plumber, not the domain expert.
- (3) **The Microsoft** owns the loop OS layer, the standard interface everyone writes to.
- (4) **The Google** indexes grounded data across domains: the capability marketplace where one finds a loop that already exists.
- (5) **The Apple** integrates the loop into a consumer product so cleanly that no one thinks about the loop at all.

Justification. By Definition 5.1, Transition 2 requires an interoperable protocol (the Microsoft position), a substrate for it (the Cisco position), an index of the ecosystem it enables (the Google position), and a consumer-facing integration of it (the Apple position). By the standard-defined character of Transition 2, the pre-standard incumbents are precisely those with proprietary local dominance (the Novell position). $\square$

Example 5.4 (Novell). Novell dominated local area networks because it solved file and print sharing for proprietary local networks. It did not own the internet. When TCP/IP arrived, Novell's value collapsed. Its lesson: proprietary local dominance is fragile if interoperability wins.

Remark 5.5 (The Framework's Response Is Precise). If loop closure standardizes, value migrates to whoever owns deployment channels or grounded data at scale (the flagged failure mode of Application Paper I). The moat is not the protocol; the moat is the grounded data. Novell had a proprietary protocol but did not own proprietary data at scale. It lost. The integrators who own grounded data at scale do not lose, because grounding is not a protocol; it is an asset.

Theorem 5.6 (The Novell Warning). *If loop standardization happens, a domain integrator whose moat is a proprietary loop protocol is stripped of the moat; a domain integrator whose moat is grounded data at scale is not.*

Proof. By Theorem 4.4(2), grounded data is non-derivable and non-purchasable. Standardization is a standardization of *interfaces*, not of *assets*; it renders proprietary interfaces commodities (Theorem 2.5, applied at the loop-protocol layer) while leaving the asset classes untouched. The protocol-only integrator therefore holds only a commodity after standardization; the grounded-data integrator holds the asset. $\square$

Corollary 5.7 (The Strategic Correction). *Under the phone path, the strategic advice changes in three respects and in no other:*

- (1) *Do not build a proprietary loop protocol as your moat. Build it, use it, but design for the standard; be ready to adopt the standard when it arrives.*
- (2) *Own the grounded data. It survives standardization; it is the asset that cannot be commoditized.*
- (3) *Be the Novell that becomes Cisco, Microsoft, or Google. Do not be the Novell that stays Novell.*

Proof. Items (1) and (3) follow from Theorem 5.6: protocols are commoditized by standardization, so the protocol-owner must be ready to occupy the standard-era positions of Proposition 5.3. Item (2) follows from Theorem 4.4(2): grounded data is the only asset that survives standardization. $\square$

6 Strategy: The Fork

The framework's advice to a founder is: close the loop, log the data, own the domain. Under the phone path, one line is added: design for the standard, and own the data that survives it. The two paths—staggered and standardized—are alternatives with a common core. This section states the fork and the hedge.

Proposition 6.1 (The Advice That Survives Either Path). *The advice that the moat is the logged experience; that the agent should log everything; that the integrator should resist selling the loop; that the firm should stay quiet because incumbents are distracted by layer-directed strategy; and that loop-directed strategy is aligned with the value layer, survives both the staggered path and the standardized path.*

Proof. By Theorem 4.4, the value-layer shift, grounded-data propriety, and vigilance against layer-directed strategy are invariant under P. Each listed item is a strategic consequence of those invariants, given in Application Paper I and Application Paper II. Neither the staggered nor the standardized path negates the invariants. $\square$

Proposition 6.2 (Path Dependence of the Advice). *The advice enter narrow, expand along max T is path dependent: it is mandatory on the staggered path and unnecessary on the standardized path.*

Proof. By Application Paper II, the narrow-entry strategy is derived from Theorem 2.7 and the transfer structure of the environment landscape. On the staggered path, $T > 1$ holds only within modality clusters, so a general-purpose loop is unaffordable and narrowing is required. On the standardized path, condition (P4) asserts $T > 1$ everywhere, so a general-purpose loop is affordable from the start and the narrow-entry phase can be skipped. Hence the advice holds exactly on the staggered path. $\square$

Corollary 6.3 (The One Change for a Founder). *If the loop is plug-and-play and transfer is high, a founder can start with a general-purpose loop. Everything else in the founder’s advice survives. The grounded-data moat is still the point.*

Proof. Immediate from Propositions 6.1 and 6.2, and Theorem 4.4(2). $\square$

Remark 6.4 (The Hedge). The framework tells the founder what to watch, not what will happen. The prescription is to build for the staggered scenario and watch for the standardized one. If loop closure begins to standardize—if the microchip moment starts in the loop layer—pivot toward the general-purpose play. If it does not, stay narrow and expand along max T . The two-plays strategy follows from the two-branches structure.

7 The Revised Prediction

Conditioning on the analysis above, the phone scenario updates the framework’s timeline predictions. What follows is not a framework theorem; it is a framework-informed read. We state it as a prediction and mark its status.

Prediction 7.1 (Revised Timeline Under Partial Standardization). (1) **Specialized humanoids**

reach phone prices first. Cleaning, cooking, laundry, and eldercare each close a domain-specific loop and each reach consumer-electronics price by 2035–2040.

(2) **General-purpose humanoids remain expensive.** Not because hardware is expensive, but because closing loops across many domains remains hard. The general-purpose robot is a 2050+ product.

(3) **The staggered eras survive, with larger clusters.** Specialized robots arrive first; general-purpose robots later. The gaps between eras are narrower than in the default scenario, but they exist.

- (4) **The integrator moat survives.** Whoever owns the loop in each specialized domain still owns the grounded data. Consumer-electronics pricing does not change this.
- (5) **The value layer still shifts.** Grounding is still the moat, even in the consumer scenario.

Remark 7.2 (Status of the Revised Prediction). Prediction 7.1 is an empirical read conditioned on the weights the framework assigns to partial standardization. It is not a theorem. The framework grants that hardware commoditization and automatic calibration are likely, that full loop standardization is difficult, and that transfer improves within modality clusters but stays below 1 across them. Under those weights, the revised prediction follows. Under different weights—for example, a full solution to domain-general grounding—the consumer scenario plays out completely, and the framework’s industrial predictions need substantial revision. The framework is honest about this conditionality.

Prediction 7.3 (Consumer Humanoids Are Not General-Purpose Yet). A \$1,000 robot that cleans a kitchen, loads a dishwasher, and folds laundry is plausible by 2040. A \$1,000 robot that also fixes a car, tends a garden, and repairs a circuit board is not. The consumer-electronics scenario is real for specialized domains; it does not extend to general-purpose.

Proposition 7.4 (The Honest Consequent). *The framework is not wrong if the phone scenario plays out. It is a conditional theory whose antecedent turned out to be false. The core insight (grounding is the moat) survives; the specific predictions about timing and industrial shape do not.*

Proof. By Proposition 3.5, the phone scenario negates the antecedent IC on which the timeline predictions rest (Theorem 4.1). By Theorem 4.4, the core structural claims do not rest on that antecedent. Hence the framework’s failure under P is a failure of its conditionals, not of its axioms: the consequent follows from the antecedent, and the antecedent is what P negates. $\square$ $\square$

8 Open Problems

Open Problem 8.1 (Loop Standardization Certificate). What would a loop-standardization certificate even be? The framework says sensing, prediction, falsification, revision, and persistence can be standardized as an interface, but the domain-specific calibration cannot. Can this separation be made precise enough to test?

Open Problem 8.2 (The Two Transitions Distinguishability). Can the loop-layer microchip moment and the loop-layer internet moment be observationally distinguished in advance? Each transition has a distinct signature (convergence of a reusable stack vs. emergence of an interoperable protocol), and the founder’s hedge depends on detecting the first before the second.

Open Problem 8.3 (Transfer Threshold for Standardization). What is the threshold value of T above which loop standardization becomes self-reinforcing? The revised prediction assumes T stays above 1 only within modality clusters; the phone scenario requires $T > 1$ almost everywhere. The regime transition between these is not resolved.

Open Problem 8.4 (Standardization Rescue of the Model Layer). If loop standardization commoditizes grounded data interfaces, does any value return to the model layer, or does the value remain at the loop layer with the interface layer as a gatekeeper (the Microsoft position of Proposition 5.3)?

9 Conclusion

The phone scenario is the sharpest challenge the framework faces, and it is worth taking seriously. It attacks the one variable the industrial argument rests on: integration cost. We have formalized the scenario as four conditions, separated the two historical transitions it compresses (the microchip moment and the internet moment in the loop layer), identified the present as the Novell era of robotics on the phone path, and derived the strategic correction—design for the standard, own the data that survives it.

The results:

1. The phone scenario attacks the anchoring claim IC (Proposition 3.5).
2. Under P, four timeline predictions are falsified (Theorem 4.1) and one is weakened (Corollary 4.2). This is the anticipated “Integration Standardization” failure mode (Remark 4.3).
3. Four core claims are invariant under P: the value layer still shifts, grounded data is still non-purchasable, the winner is still unpredictable, and the model layer still commoditizes (Theorem 4.4). Grounding is the moat, not the model; this survives the counterfactual (Corollary 4.5).
4. The two transitions are the loop-on-a-chip and the interoperable loop protocol; we have neither, and are therefore in the Novell era (Proposition 5.2).
5. Standardization commoditizes protocols, not assets; the grounded-data integrator survives, the protocol-only integrator does not (Theorem 5.6).
6. The strategy changes in exactly one respect on the phone path: the narrow-entry phase can be skipped (Propositions 6.1, 6.2); otherwise the advice survives (Corollary 6.3).
7. The revised prediction: specialized humanoids reach phone prices first, general-purpose humanoids remain a late product, staggered eras survive with larger clusters, the integrator moat and the value-layer shift survive (Prediction 7.1).
8. The framework is honest about its conditionals: if its antecedent is false, its consequents are false too, and the failure is a failure of the condition, not of the core insight (Proposition 7.4).

The one-sentence version: if humanoids cost like phones, several of the framework’s timeline predictions are wrong, but its core claim—that grounding is the moat—survives; the era arrives faster and more unified, but the winners are still the loop-closers, and the model layer still commoditizes.

The honest uncertainty should stand last. If someone solves domain-general grounding—if the loop becomes genuinely plug-and-play across environments—the consumer scenario plays out fully and the framework’s industrial predictions need substantial revision. The framework’s central claim is what makes that problem hard: grounding is lived, not purchased. The framework tells you what to watch. It does not tell you what will happen.

A Summary of Definitions

Symbol	Name	Reference
$G(\mathcal{A})$	Grounding coefficient	Paper 0, <i>The Inference-First Picture of Mind, Logic, and Mathematics</i>
$G_{\mathcal{E}}(\mathcal{A})$	Environment-indexed grounding	Paper 0, <i>The Inference-First Picture of Mind, Logic, and Mathematics</i>
P	Phone scenario (four conditions)	Definition 3.1
c_{hw}, c_{env}, c_{int}	Component, environment, integration cost	Application Paper I
$T(\mathcal{L}, \mathcal{E}')$	Transfer coefficient	Application Paper II
IC, INT, LS, TK, IBM, ENT, T, INV	Prediction shorthands	Section 2

B Summary of Theorems

Theorem	Statement
3.3	Conditions (P1)–(P2) are engineering problems
3.4	Conditions (P3)–(P4) are framework challenges
3.5	The scenario attacks IC
4.1	Four timeline predictions falsified under P
4.2	Integrator prediction weakened
4.4	Four core claims invariant under P
4.5	Grounding is the moat; survives the counterfactual
5.2	We have neither transition: the Novell era
5.3	Who rules each gate of the standardized loop layer
5.6	The Novell warning: protocols commoditize, data does not
5.7	Design for the standard; own the data that survives it
6.1	The advice that survives either path
6.2	Narrow entry is path dependent
6.3	The one change for a founder
7.4	Failure is of the condition, not the core insight

C A Note on Method

This paper is a stress test of a conditional theory. Its formal results show which claims depend on which antecedents; they do not predict which antecedent is realized. The revised prediction of Section 7 is an informed read with stated weights, separated from the theorems. The distinction is the point: a structural theory that could not be stressed would not be a theory, and a stress test that dishonestly shielded its target would confirm nothing.